Analytics to Action

Datafication

(+ a few announcements)

Anders Koed Madsen, February, 2026

A few announcements before we start

You have now been allocated a group and assigned a case.

a. Not all of you have gotten your preferred case - but we are sure that you'll learn much from the case you got.
b. It is OK for two groups to swap cases if you inform us.
c. The groups are set and cannot be redone

You will receive access to the data during this week

a. We will most likely just share the the data on OneDrive and leave it to you to download it (through a link on learn)
b. Make sure to treat the data with care. Do not upload to cloud based systems that are not on DTU servers.
c. Please fill in this survey, so we can get a sense of the data infrastructure that you plan using

Theme 2: Making Data Valuable

How to make data and data analysis valuable to different stakeholders in practice.

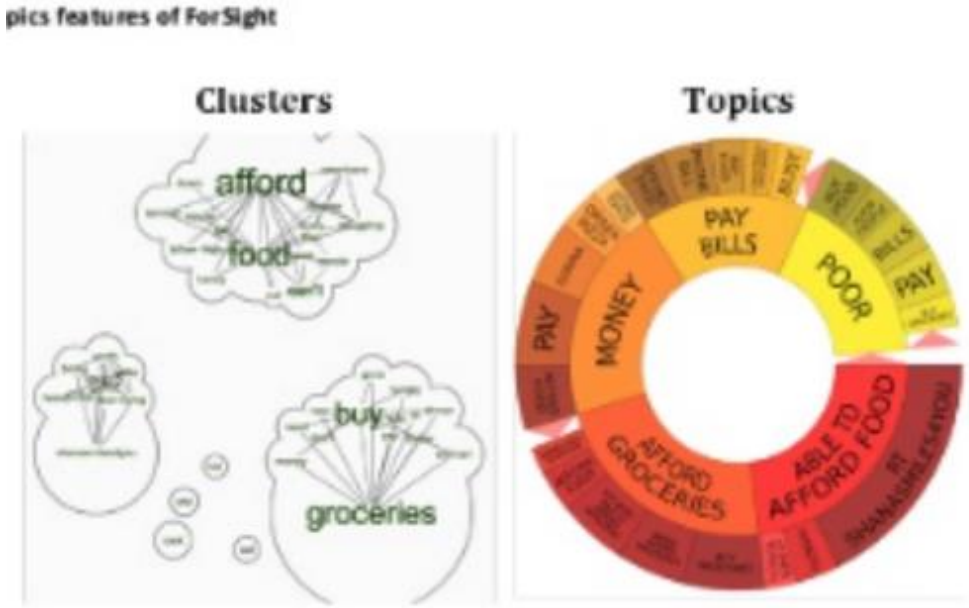
Three sessions

1. **Datafication**
2. Exploratory data analysis
3. Participatory data design

Readings for today

- Mejias, U. A., & Couldry, N. (2019). Datafication.
- Flyverbom, M., & Madsen, A. (2015). Sorting data out.

Value requires ‘fit’ with context



Brøndby fjerner omstridt coronakort efter kritik fra styrelse

Efter kritik fra Styrelsen for Patientsikkerhed har man nu besluttet at fjerne det omstridte kort.

Agenda for this lecture

- Datafication: what is it and why does it matter?
- The four ‘analytical moments’ from Flyverbom & Madsen
- An example of datafication in ECHOLAB
- A prompt for your case work

Data is the "material produced by abstracting the world into categories, measures and other representational forms [...] that constitute the building blocks from which information and knowledge are created" (Kitchin, 2014)

"**Datafication**" implies that something is made into data" (Mejias & Couldry, p.2)

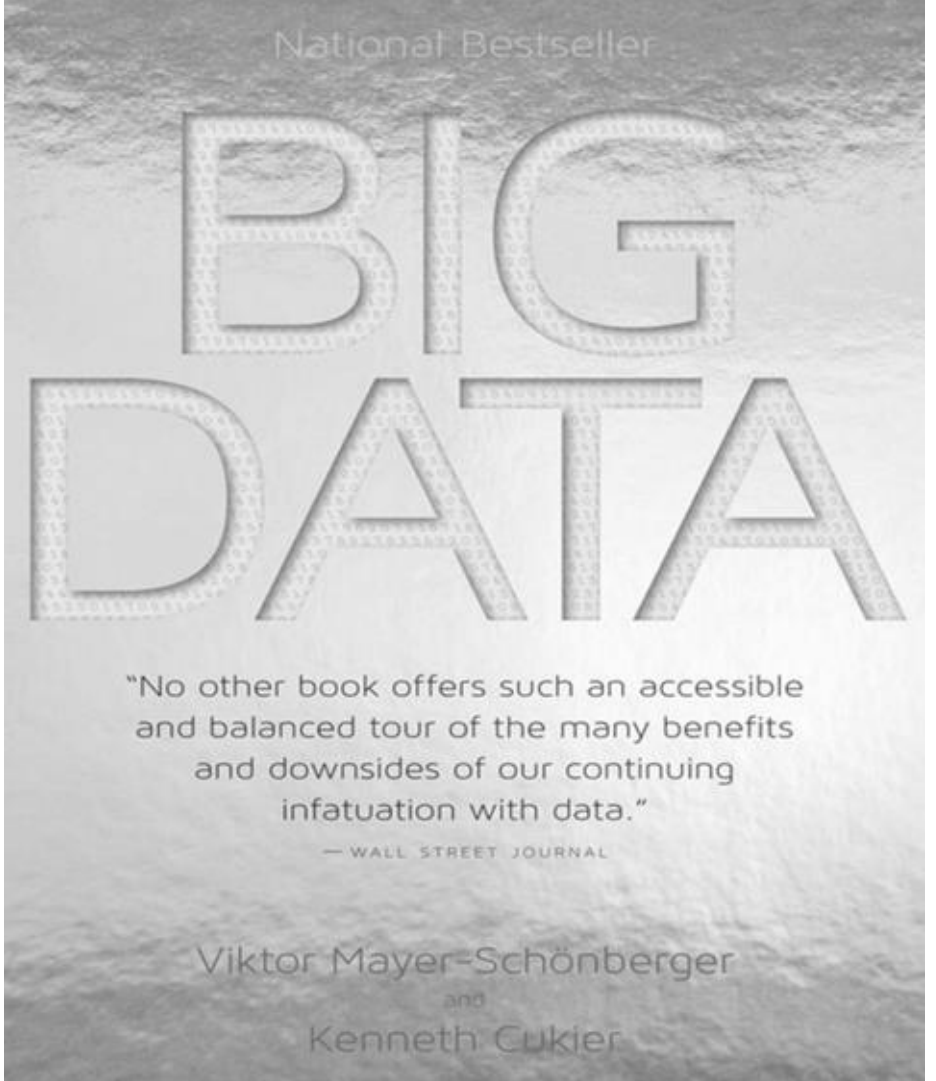
- it's a verb!

Datafication vs. Digitization

(From Cukier & Meyer-Schönberger, 2013)

"**Data** refers to a description of something that allows it to be recorded, analysed and reorganized"

"This is very different from **digitization**, the process of converting analog information into the zeros and ones of binary code"

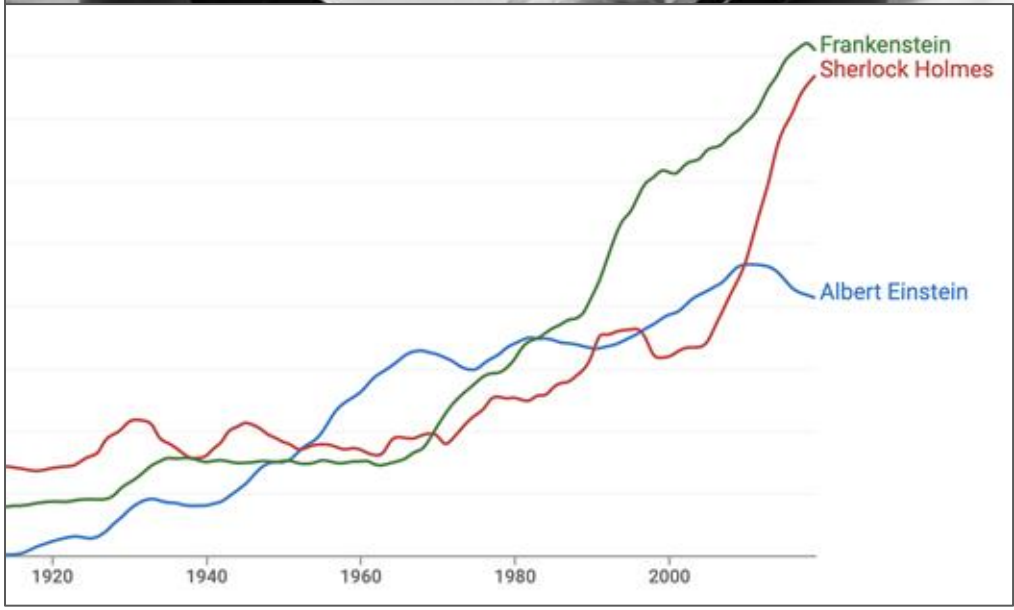


Datafication vs. Digitization

(From Cukier & Meyer-Schönberger, 2013)

"First, Google digitized text: every page was scanned and [...] transformed into a digital copy [...] that only humans could transform into useful information - by reading".

"Google wanted more. The company [...] used optical character recognition software that could take a digital image and recognize the letters, words, sentences, and paragraphs on it. The result was <u>datafied text</u> rather than a digitized picture of a page".



Datafication vs. Digitization

(From Cukier & Meyer-Schönberger, 2013)

billetto.

- Many organization lack a focus on datafication when they make strategic plans for digitization.
- The data bi-effects of datafication are not thought through.
- Copenhagen Municiplaity's ticketing system as a good example

KØBENHAVNS KOMMUNE

“Digitalization turbocharges datafication. But it is not a substitute.”
(Cukier & Schönberger, 2013)

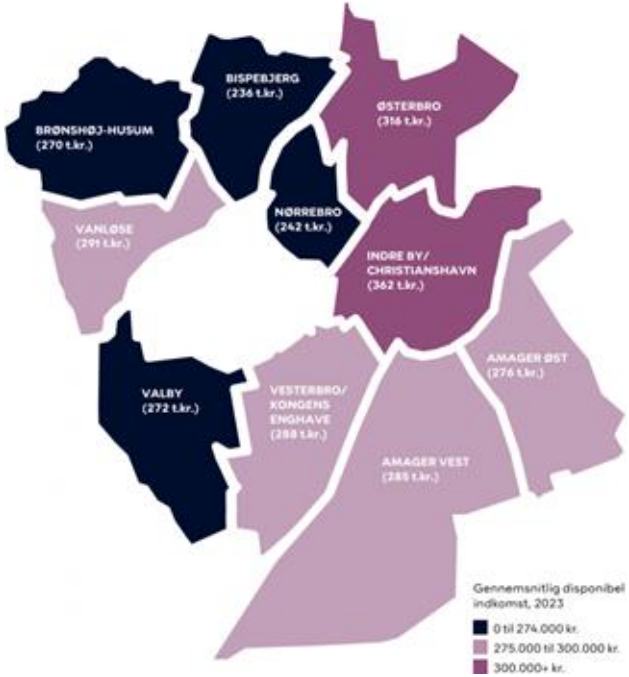
Datafication practices have long histories
(case: urban statistics)

"there went out a decree from Caesar Augustus, that all the world should be taxed".

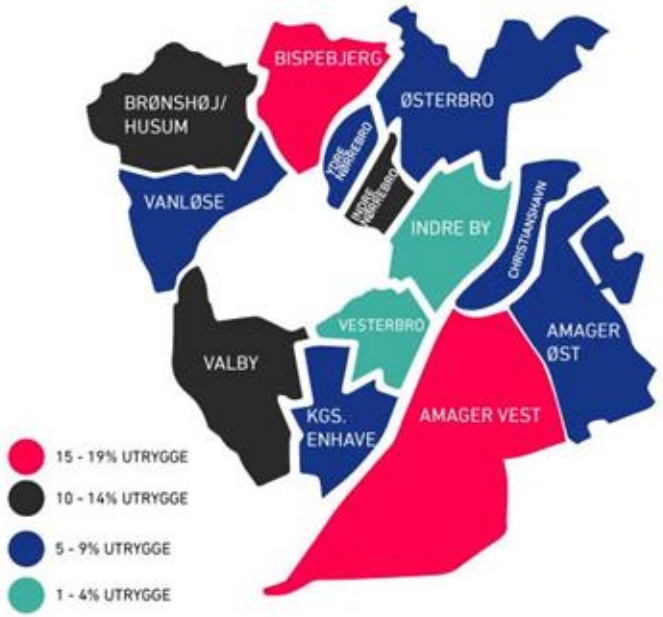
Datafication practices have long histories
(case: urban statistics)

Napoleon's cadastral maps

Datafication practices have long histories
(case: urban statistics)



Foundation for taxation



….but also feelings of safety!!

Turn to your neighbor for 3 minutes

Can you think of contemporary datafication practices still bear traces of pre-digital ways of measuring?

If yes, how and with what consequences?

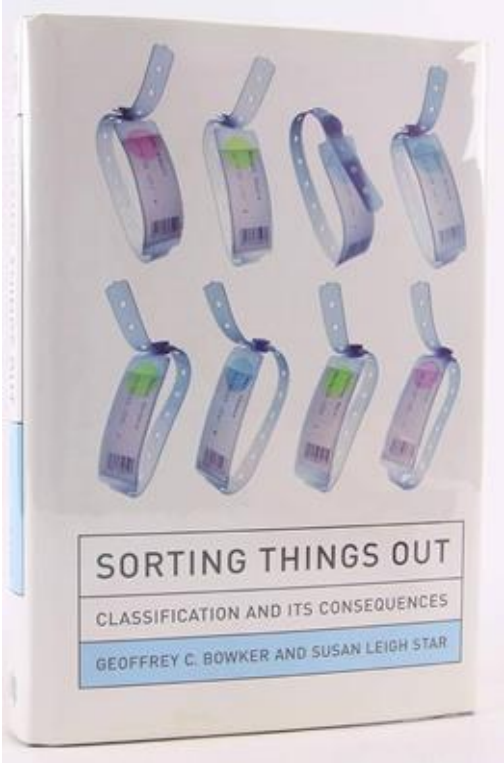
“Raw data” is both an oxymoron and a bad idea

◄ Geoffrey Bowker, Professor of Information Science, UC Irvine.

Studies classification systems and their consequences ►

Oxymoronic, because how does data become data?

Bad idea, because what if we pretended it was without consequences?



Datafication battles are social battles

In terms of actors, we have corporations, states, and various civic (activists, journalists, etc.) and even non-state (terrorists, hackers) actors, all of which can produce, collect and analyse data for different purposes.

(Mejias & Couldry, p.2)

Datafication battles are social battles

Steven Epstein (1995) The Construction of Lay Expertise: AIDS Activism and the Forging of Credibility in the Reform of Clinical Trials

Lury & Gross (2014): THE DOWNS AND UPS OF THE CONSUMER PRICE INDEX IN ARGENTINA

…that sometimes exist inside an organization!
(case: research quality)

To management can enforce specific metrics as incentives

TEKNOANTROPOLOGISK LABORATORIUM PÅ AAU HÆDRET MED INTERNATIONAL PRIS

But people 'on the floor' (in this case researchers) will invent their own metrics (e.g. prizes)

…while digital actors are also in the business of entering this conversation

"The production of data cannot be separated from two essential elements: the **external infrastructure** via which it is collected, processed and stored, and **processes of value generation** (Mejias & Couldry, p.2)"

Turn to your neighbor for 3 minutes

Can you think of datafication battles in today's society? Where people are in disagreement about how to turn the world into data and indexes?

If yes, which actors are active in proposing different strategies for producing, collecting and analysing data?

How to think about datafication in a world where data is abundant?

Flyverbom & Madsen

By Brad Brown, Michael Chui, and James Manyika

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Radical customization, constant experimentation, and novel business models will be new hallmarks of competition as companies capture and analyze huge volumes of data. Here's what you should know.

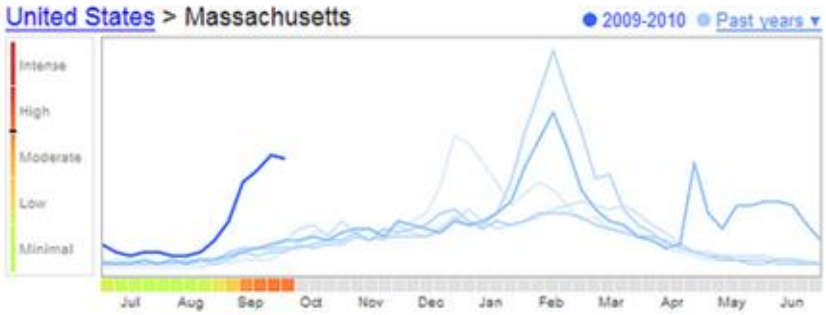
The Economist
MAY 6TH–12TH 2017
Crunch time in France
Ten years on: banking after the crisis
South Korea's unfinished revolution
Biology, but without the cells
The world's most valuable resource
Data and the new rules of competition
Google
amazon
UBER
Google
TESLA

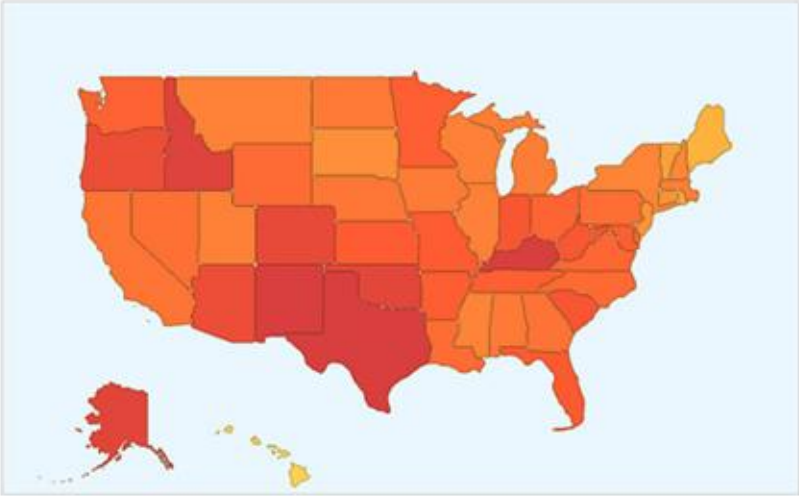
CHRIS ANDERSON SCIENCE JUN 23, 2008 12:00 PM

The End of Theory: The Data Deluge Makes the Scientific Method Obsolete

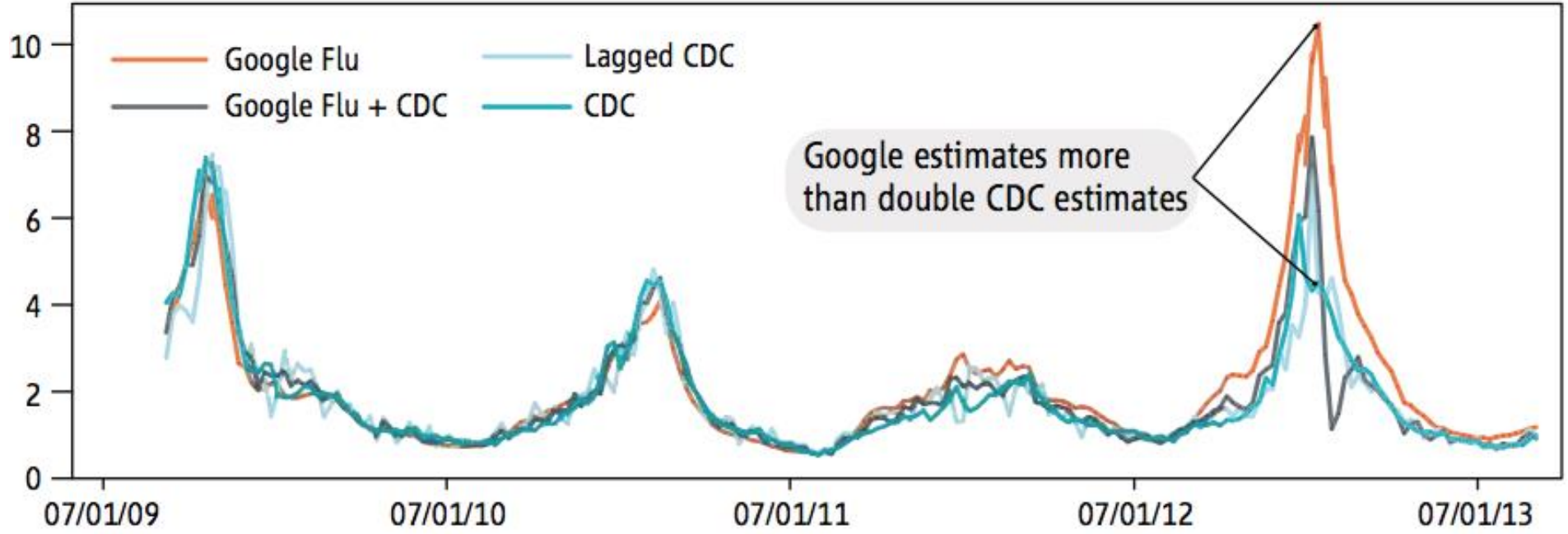
Illustration: Marian Bantjes "All models are wrong, but some are useful." So proclaimed statistician George Box 30 years ago, and he was right. But what choice did we have? Only models, from cosmological equations to theories of human behavior, seemed to be able to consistently, if imperfectly, explain the world around us. Until now. Today companies [...]

…but this hides how ‘big data’ always rests on consequential datafication choices





Lazer, D., Kennedy, R., King, G., & Vespignani, A. (2014). The parable of Google Flu: traps in big data analysis



Lazer, D., Kennedy, R., King, G., & Vespignani, A. (2014). The parable of Google Flu: traps in big data analysis

Four ‘analytical moments’ in datafication

I.e., situations where it makes sense to think about datafication consequences (Flyverbom & Madsen)

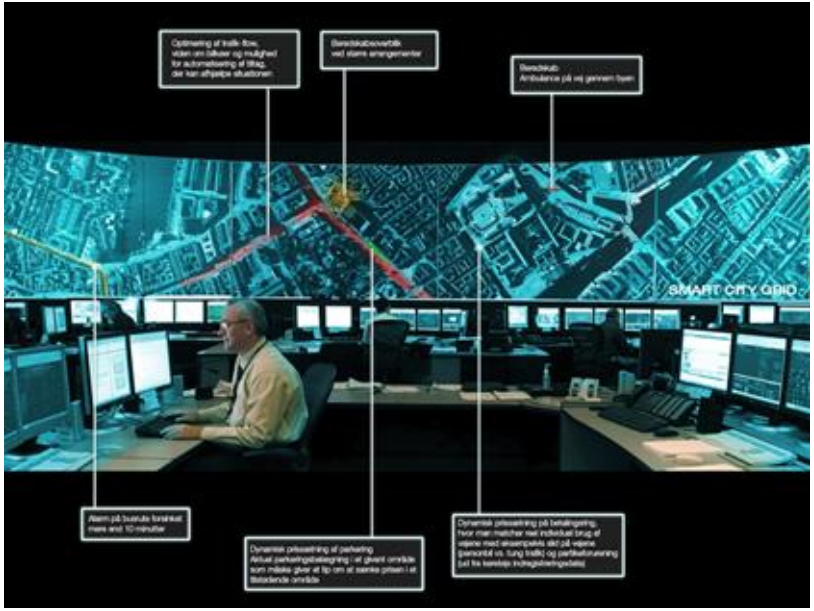
Production: Where human conduct and movements of objects are translated into a stream of data that can be stored and processed by a computer

Structuring: Choices about the databases, classification systems and metadata through which data is ordered and readied for systematic analysis.

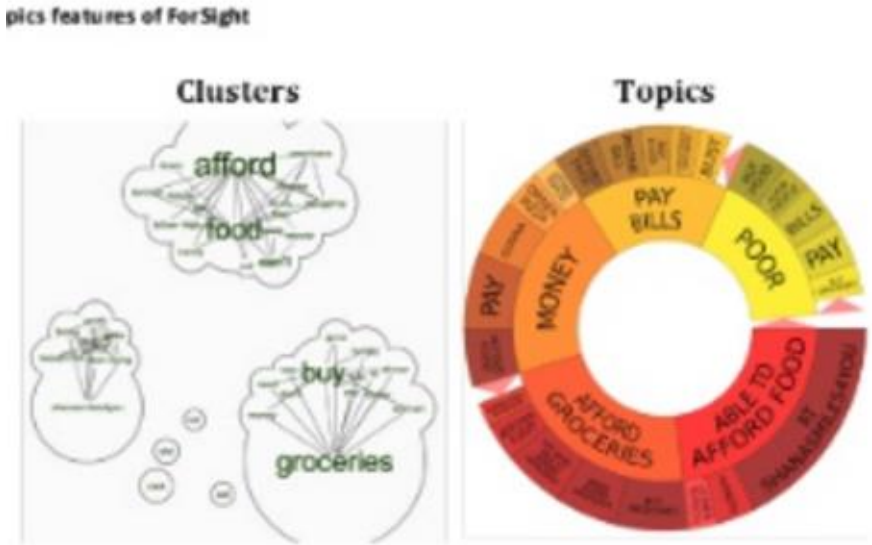
Distribution: The way access to databases and distribution of digital traces are negotiated between data-owners and end-users

Visualization: Turning the available data into visuali-ations that give insights into the aspect of the world that one is interested in

COPENHAGEN
SOLUTIONS LAB



UN GLOBAL PULSE



	Copenhagen Solutions Lab		UN Global Pulse	
	Technology	**Context**	**Technology**	**Context**
Production	Sensors, RFID tags, Wi-fi signals	Ongoing infrastructure projects Municipal strategies	SoMe Platforms & API's and data-brokering	Re-design of interfaces and data end points
Structuring				
Distribution				
Visualization				

Turn to your neighbor for 3 minutes

Think about the two cases we are working with in this course - Rigshospitalet and Will & Agency.

Can you note down important questions regarding the production of data that we should keep in mind?

	Copenhagen Solutions Lab		**UN Global Pulse**	
	Technology	Context	Technology	Context
Production				
Structuring	Municipal database w/ standardized metadata across subunits	Decision about metadata and internal power struggles.	Data categories driven by platform desire to produce clicks	User-driven metadata in constant development
Distribution				
Visualization				

Turn to your neighbor for 3 minutes

Think about the two cases we are working with in this course - Rigshospitalet and Will & Agency.

Can you note down important questions regarding the structuring of data that we should keep in mind?

	Copenhagen Solutions Lab		**UN Global Pulse**	
	Technology	Context	Technology	Context
Production				
Structuring				
Distribution	Open database for municipal stakeholders	Interest-conflicts connected to procurement and branding	Communication from GP to the broader UN strategy team	Diverging views on criteria for legitimate data
Visualization				

	Copenhagen Solutions Lab		**UN Global Pulse**	
	Technology	Context	Technology	Context
Production				
Structuring				
Distribution				
Visualization	Real-time data on geo-mapping software	Continues a tradition dating back to cadastral maps	Real time indicators based on semantic patterns	Invention of baseline for anomaly detection

Turn to your neighbor for 3 minutes

Think about the two cases we are working with in this course - Rigshospitalet and Will & Agency.

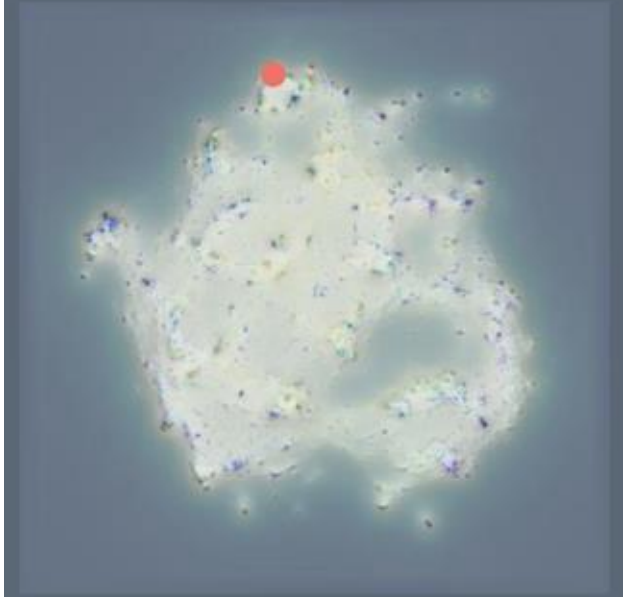
Can you note down important questions regarding the distribution/visualization of data that we should keep in mind?

Example

Grounding the technical discourse on AI for public engagement at a museum

#3303

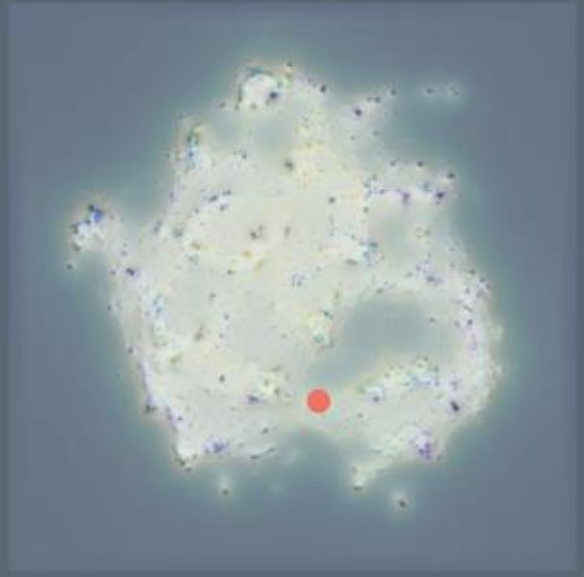
Secure spectrum sensing



Secure spectrum sensing: Recent advances in cognitive radio networks have led to the development of algorithms that can defend against attacks and improve the efficiency of cooperative spectrum sensing. These algorithms analyze the strategies of attackers, verify sensory data, and calculate thresholds for classification accuracy. They also mitigate the influence of malicious users and improve the reporting mechanism for estimating primary traffic. By using these algorithms, researchers aim to maximize the throughput of cognitive radio networks and achieve secure and efficient spectrum sensing.

#1793

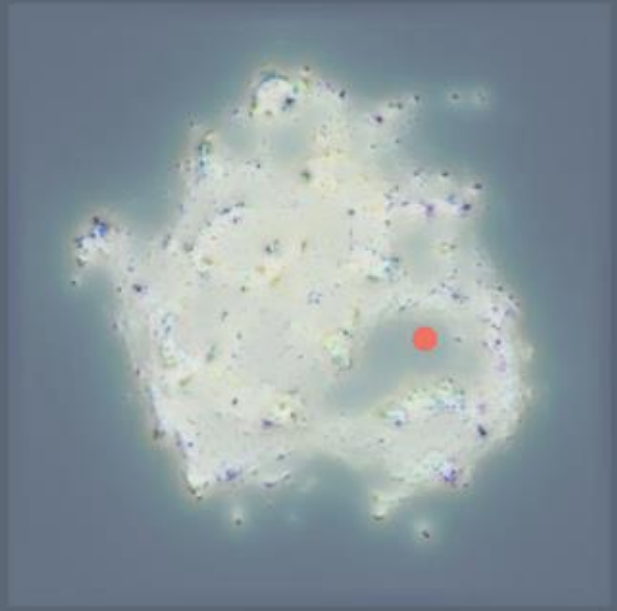
Chicken diet optimization



Chicken diet optimization: Researchers have been using various algorithms, such as artificial neural network models and optimization algorithms, to analyze the responses of broiler chickens to different diets and nutrients. These algorithms help in estimating early egg production, nutritional requirements, and optimizing dietary components for better performance. Additionally, other algorithms like Bayesian networks and Gaussian process regression are being used to predict the metabolizable energy and nutrient intake of broiler chickens. These findings contribute to improving the understanding of how to optimize the diets of broiler chickens for better growth and performance.

#1574

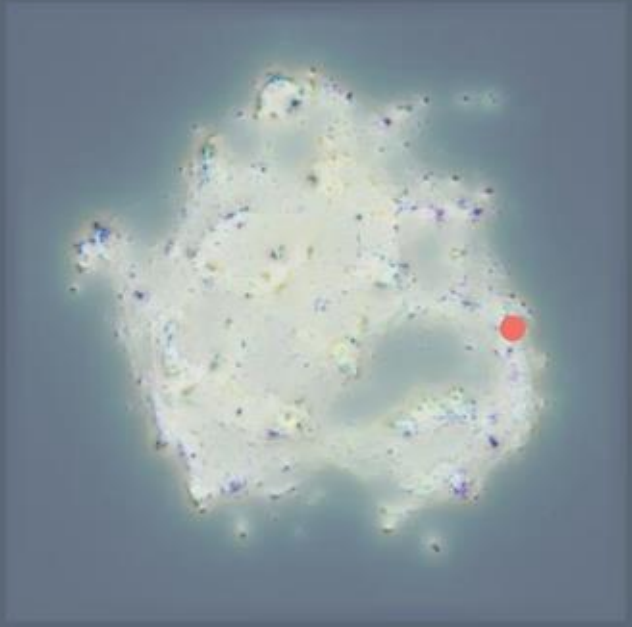
Food image analysis



Food image analysis: Researchers have been exploring the capabilities of algorithms in recognizing and analyzing food images. They have found that deep convolutional neural networks, image processing, computational intelligence, and deep learning algorithms can accurately identify dish types, food ingredients, cooking methods, and even estimate nutrient content and calorie values. These algorithms have the potential to be used in dietary assessment, food category recognition, and other food analysis applications.

#2527

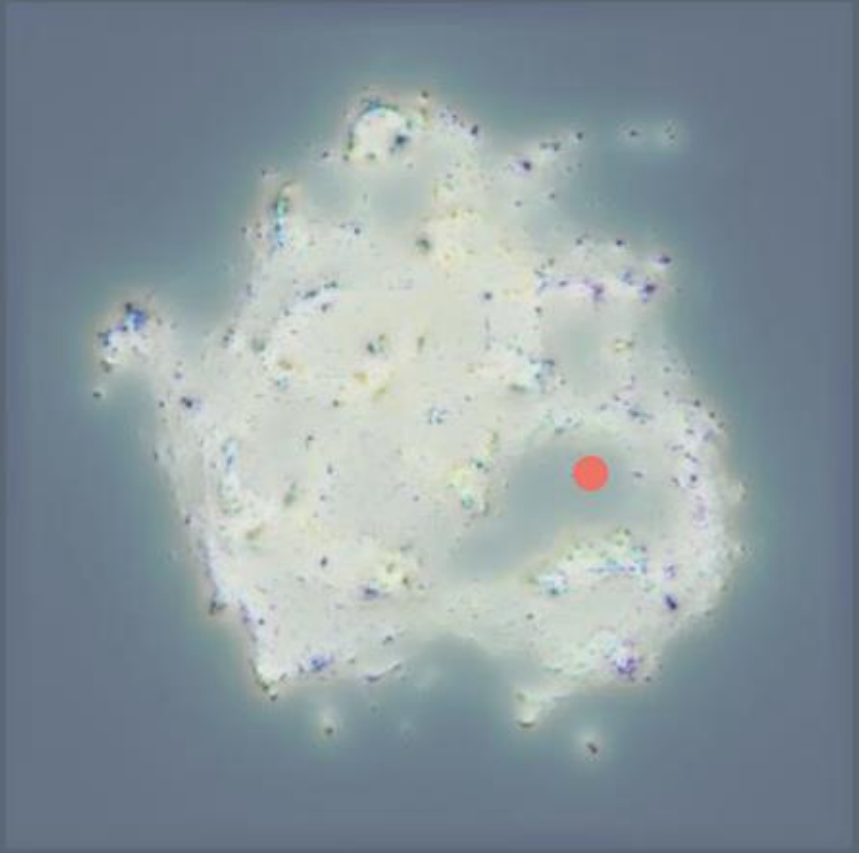
Dietary change algorithms



Dietary change algorithms: Researchers are exploring how algorithms can be used to understand and classify different stages of dietary change. They are developing algorithms to determine a person's stage of change for factors like fruit and vegetable consumption, dietary fat intake, and low-fat eating. These algorithms can help identify cognitive profiles, psychosocial variables, and readiness to change behavior, providing valuable insights for promoting healthier eating habits.

#1574

Food image analysis



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Get some synthetic opinions on this topic

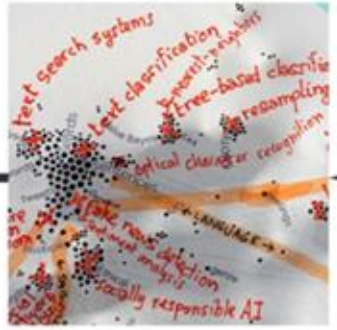
1M Scopus Documents
(abstract + metadata)

Extract named entities
and keywords
= expressions

Network of 7.6K expressions
connected by co-occurrence
in Scopus documents (85K
edges)

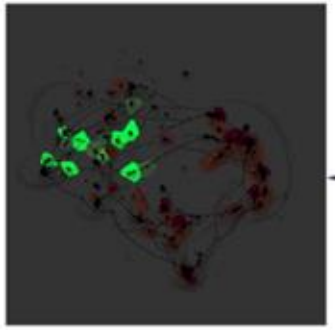
Detect 166 clusters of
expressions through
clique percolation

Extract a sample of
documents representative
of each cluster, and
read them.



Manually annotate the
network layout with
a summary of each cluster

Network annotated and
visualized as a map



Statistical analyses
of the prevalence of
search terms in different
areas of the network map

Challenge

Spot the datafication issue at the level of production

Prompts for your case work and presentations

- How was your data made and how is it now made available to you? Could it be otherwise?
- Who made it and with what interests? Do these interests align with your interests? Could there be other interests in the company/organization?
- What would need to change on the level of production, structuring, and/or distribution for you to best address the challenge you have been given by your case company? And how realistic are those changes?
- What can be done within the frame of the available datafication? And with what reservations vis-a-vis the results?